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Physics-informed Fourier neural network with self-adaptive loss weight for modeling transient wave propagation

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ABSTRACT

The forward modeling and parametric inversion of mechanical waves in physical systems are of significant importance in a multitude of scientific and engineering disciplines. Physics-informed neural networks (PINNs), which have the potential to seamlessly integrate physical models and observational data, have emerged as an effective tool for wavefield modeling and inversion. Nevertheless, when solving for wavefields with multi-frequency and high-frequency characteristics, the inherent spectral bias and loss convergence imbalance of PINNs ultimately result in their inability to accurately model wave propagation. This paper proposes a new framework, the Physics-informed Fourier neural network (PIFNN), to address these issues by incorporating Fourier feature mapping and adaptive loss weights based on the Neural Tangent Kernel (NTK) theory. The robustness and effectiveness of the PIFNN method are demonstrated by improved accuracy in simulating wave propagation in numerical experiment. Furthermore, the training dynamics of wave propagation simulated by PIFNN are analyzed through the lens of NTK to enhance interpretability.

1. Introduction

The forward simulation of the mechanical wave in physical systems, coupled with the inverse identification of their parameters, represents a significant area of research, with applications in various fields, including non-destructive testing (Yin and Shi, 2022), medical imaging (Yin et al., 2024), metamaterial structural design (Cho et al., 2024), seismology (Ding et al., 2023), electromagnetic (Jeong et al., 2024) and ocean sonar detection (Ren et al., 2022). Ultrasound, as a type of mechanical wave, is used as an effective physical sensing method, and its propagation properties in a medium including scattering, reflection and refraction can be analyzed to characterize unknown material properties and geometries (Wu et al., 2023a; Xu et al., 2024). In recent decades, numerous numerical methods have been developed to solve partial differential equations (PDEs) describing wave propagation, thereby facilitating detailed analyses of mechanical wave in diverse physical systems, including finite element method (FEM) (Li et al., 2024), finite difference time domain method (FDTD) (Shivaprasad et al., 2018), and boundary element method (BEM) (Shieh et al., 2018). Nevertheless, predictive modeling of complex spatiotemporal dynamics using traditional numerical methods can pose significant challenges in many practical scenarios: (1) The computational paradigms employed are based on mesh partitioning, necessitating the use of finer meshes and time steps for simulating ultrasound propagation in complex materials, thereby increasing the demand for computational resources (2) Numerical simulations are usually time-consuming, rendering them impractical for many applications that require repeated forward modeling, such as inverse problems, optimization design, and uncertainty quantification. (3) Traditional numerical methods are limited to integrating multi-physics phenomena and deriving equivalent simplified mathematical formulations for ease of implementation. While some methods have been developed to alleviate these issues, more cost-effective solutions are still needed (Gu and Jing, 2018).

Recently, researchers have begun to explore the application of deep neural network methods to overcome the above limitations, prompting further work on numerical simulation of ultrasound, leading to a deeper understanding of propagation mechanisms and advances in ultrasound imaging techniques. Alguacil et al. constructed a convolutional neural network with multi-scale capabilities, using data from Lattice-Boltzmann to accelerate the acoustic wave propagation simulation in a two-dimensional quiescent medium (Alguacil et al., 2021). Gantal used simulation data to train deep neural networks to learn elastic wave

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propagation from multiple point sources (Gantala and Balasubramaniam, 2022). These data-driven methods are almost 30 times more efficient than numerical methods regarding calculation speed. More importantly, in larger solution domains, more complex excitation sources and sound velocity distributions can be obtained in addition to the training set of generalization.

Despite advancements, deep neural networks remain heavily dependent on sparse data and often lack physical interpretability, limiting their applicability in many science and engineering fields. Researchers have proposed integrating physical laws with data-driven concepts to address this limitation, paving a new paradigm for scientific discovery and translation (Karniadakis et al., 2021). Within this physics-enhanced machine learning framework, different types of biases can guide learning and improve generalization even with limited data (Cicirello, 2024). Observational bias leverages informative data reflecting the underlying physics, while learning bias shapes the training process through model design and optimization strategies. Inductive bias imposes prior assumptions or physical constraints such as conservation laws and boundary conditions, and model form bias incorporates partial knowledge of the governing equations to constrain the solution space. Physics-informed neural networks (PINNs) are a representative example, where inductive and model form biases are embedded in the loss function via automatic differentiation (Paszke et al., 2017), demonstrating significant potential for solving partial differential equations (PDEs) and inverse problems, allowing diverse types of boundary conditions (BCs) and initial conditions (ICs) (Raissi et al., 2019). Karniadakis presented a first-time unified framework applicable to wave propagation and full waveform inversion that predicts the displacement field over the entire computational domain using only two time slices of wavefield data (Rasht-Behesht et al., 2022). Liu et al. applied PINNs to the numerical simulation of transcranial ultrasound, illustrating PINNs' scalability without the need for expensive solutions (Wang et al., 2023). Thakur proposed a neural green-distributed point source method for modeling ultrasonic field emission near a fluid-solid interface, which utilizes PINNs to evaluate the Green's function and its gradient, thereby addressing the shortcomings of traditional numerical methods (Thakur et al., 2024). Schmid demonstrates that PINN can accurately predict the sound pressure field and estimate the boundary conductance well, even when dealing with sparse and noisy experimental sound pressure data (Schmid et al., 2024). Moreover, migration learning enables PINNs to address diverse acoustic scattering issues and predict pertinent physical quantities using trained models, whereas conventional numerical techniques necessitate the reconstruction of the simulation (Veerababu and Ghosh, 2024). In contrast to conventional neural networks, PINNs utilize physical laws to provide prior knowledge, thereby markedly reducing the reliance on observational data while exhibiting accelerated convergence and superior generalization capabilities.

However, PINNs train pathologically or even fail completely on some problems, especially since the PDE solutions include high-frequency and multi-frequency features. Recent studies have revealed two fundamental weaknesses in PINNs (Wang et al., 2021). The first is related to the intrinsic property present in neural networks, which cannot learn high-frequency functions well, leading to inaccurate predictions, called spectral bias (Rahaman et al., 2019). Wang et al. employ the recently developed Neural Tangent Kernel (NTK) theory to confirm that PINN does indeed exhibit the phenomenon of spectral bias, which is the main reason that hinders the learning of high-frequency components (Wang et al., 2022). This presents a significant challenge for the multiscale modeling of ultrasound. This paper reveals this phenomenon in the context of a one-dimensional wave equation. The second fundamental weakness of PINNs is transforming the forward problem into an unconstrained optimization problem with multiple loss functions, which encounters an order of magnitude difference between the different terms of the loss function (Wang et al., 2022). Consequently, an imbalance in gradient flow dynamics of the different terms of the composite loss function during training progress results in the inability to obtain the correct solution, indicating that the PDE residual terms with scattering information and the BC terms with reflection information can 'fail' rendering the standard PINN unable to consistently and accurately capture the reflection and scattering features in transient mechanical wave propagation.

In light of the aforementioned analysis, one potential avenue for addressing the training pathology is to empirically provide an 'information encoding' that mitigates the difficulty of fitting known high frequencies. Tancik et al. introduced a Fourier feature network that employs a straightforward mapping of Fourier features to enhance the capacity of a fully connected network to learn high-frequency functions (Tancik et al., 2020). Jagtap et al. proposed the PINN with an adaptive activation function (aPINN), which is capable of continuously capturing the diverse frequency bands present in the solution (Jagtap et al., 2020). Another way to alleviate the training pathology is to balance the multitasking loss function through a weight adjustment strategy. McClenny proposed a soft attention mechanism approach that can solve complex problems more efficiently by subdividing the weights of each configuration point (McClenny and Braga-Neto, 2020). The empirical weight strategies limit the optimization of arbitrary ICs and BCs. Therefore, developing a dynamic adjustment strategy based on different terms of the loss function gradient flow is necessary. Xiang et al. proposed a method based on maximum likelihood estimation to set the weights of different losses, improving prediction accuracy (McClenny and Braga-Neto, 2020). In addition, wang et al. proposed a new optimization method to adaptively balance the convergence speed based on the law that neural networks converge along different decomposition directions of the eigenfunctions of the NTK (Wang et al., 2018). In particular, Lu et al. proposed a PINN method with hard constraints (hPINNs), incorporating strictly ICs and BCs into the neural network architecture by modifying the network output, eliminating the imbalance in gradient flow dynamics (Lu et al., 2021). Overall, while these improvements, including modifications to activation functions (Al-Safwan et al., 2021; Huang and Alkhalifah, 2022), enhancements to input representations (Huang and Alkhalifah, 2024), and refinements to incorporating boundary conditions (Alkhadhr and Almekkawy, 2023), have enhanced the ability of PINNs to capture high-frequency wavefield solutions. intrinsic spectral bias and loss convergence imbalance continue to limit accurate learning of high-frequency components, suggesting that further methodological developments are needed to fully address these training pathologies.

The significant challenges associated with PINNs in modelling wave propagation problems are the spectral bias, which hinders accurate learning of high frequency components critical for scattering and reflection, and the loss convergence imbalance, which prevents boundary and initial conditions from being satisfied accurately, thereby obstructing convergence to the exact solution. This study addresses the training pathology of PINNs in modeling wave propagation and provides a reliable solution to the fundamental problem in time-domain ultrasonic wave modeling. In particular, it investigates the applicability and generality of PINNs in capturing reflections and scattering generated by wave interaction with complex medium. Specifically, it focuses on the accurate detection of reflection effects induced by interlayer medium using Fourier feature PINNs, the fast convergence using an adaptive weighting strategy, and the study of the relationship between the training dynamics of the neural network and the accuracy in predicting wave propagation features. This fundamental work can be further extended to modeling wave propagation in complex medium with arbitrary velocity distributions. The main contributions and implications of the current work are summarized as follows:

- A novel framework, PIFNN, has been devised to address wave propagation issues. PIFNN integrates a Fourier feature layer and an adaptive loss weighting method based on NTK theory, thereby addressing the challenge of training pathologies arising from spectral

discrepancies and imbalance loss gradient dynamics. Compared to the PINN method, PIFNN demonstrates proficiency in capturing ultrasonic reflection and scattering features.
- The effect of different initializations of the Fourier feature mapping on the prediction performance is discussed through numerical experiments.
- The efficacy and robustness of NTK-based adaptive weighting strategies have been validated, and the strategies can be readily extended to other PINN architectures.
- The accuracy of wave propagation prediction under different network structures is evaluated using the consistent L^2 error criterion, demonstrating the superiority of PIFNN in solving the wave propagation problem.

2. Methods

2.1. Acoustic wave theory

For a linear, non-dissipative (i.e., negligible due to shear, bulk viscosity, and thermal conduction) acoustic wave, the propagation of ultrasound on an arbitrary stationary inhomogeneous medium can be described by the following equation (Stanziola et al., 2021):

$$\rho_0(\boldsymbol{x})\nabla\cdot\frac{1}{\rho_0(\boldsymbol{x})}\cdot\nabla p(\boldsymbol{x},t)-\frac{1}{c(\boldsymbol{x})^2}\frac{\partial^2 p(\boldsymbol{x},t)}{\partial t^2}=f(\boldsymbol{x},t),$$

where $\rho_0(\boldsymbol{x})$ represents the spatial distribution of the density, $p(\boldsymbol{x},t)$ describes the unknown latent acoustic pressure field, $c(\boldsymbol{x})$ and $f(\boldsymbol{x},t)$ are the wave velocity in an acoustic medium and acoustic sources. In this work, we shall focus on a simple form of Eq. (1) as flows.

$$\frac{\partial^2 u}{\partial t^2}-c^2\Delta u=f,$$

where u stands for the displacement field, Δ is the Laplace operator in the Cartesian coordinate system.

Although the wave equation is linear, solving it in complex medium is inherently challenging due to the high-frequency oscillatory, multiscale, and dispersive properties (Wang et al., 2024). These properties are exhibited by waves superimposed on time and space scales as a result of reflection, scattering, and transmission refraction induced by the junction of different velocity distributions or regional boundaries.

2.2. Conventional PINN framework

The physical phenomena of complex systems can be described by partial differential equations (PDEs). In essence, a partial differential equation on a bounded domain $\Omega\subset R^d$ can be expressed in the following general form:

$$\begin{cases}\mathscr{T}[u(\boldsymbol{x},t)]=f(\boldsymbol{x},t),\boldsymbol{x}\in\Omega,t\in[0,T]\\ \mathscr{B}[u(\boldsymbol{x},t)]=g(\boldsymbol{x},t),\boldsymbol{x}\in\partial\Omega,t\in[0,T]\\ \mathscr{H}[u(\boldsymbol{x},0)]=h(\boldsymbol{x}),\boldsymbol{x}\in\Omega\end{cases},$$

where $u(\boldsymbol{x},t)$ is the latent physical field corresponding to the PDE solution, $\boldsymbol{x}$ and are the space and time coordinates, $\mathscr{T}[\cdot]$ denotes the differential operator, $\mathscr{B}[\cdot]$ and $\mathscr{H}[\cdot]$ represent the boundary operator and initial operator.

In Conventional PINN, according to the universal approximation theorem, consider a fully connected feedforward neural network with one input layer, L hidden layers to approximate the unknown physical field u, whose input layers are $v=[x,t]$ including the spatial coordinates x and the temporal coordinates t. The number of Neurons in the kth hidden layer denotes as N_k, and the feedforward is represented as flows

$$H^1(v)=\sigma\left(w^1v^0+b^1\right),$$

$$H^k(v)=\sigma\left(w^kv^{k-1}+b^k\right),k=2,\cdots,L-1,$$

$$u_\theta(v)=w^Lv^{L-1}+b^L,$$

where $\left\{w^k,b^k\right\}$ are weights and bias of the k-th hidden layer, σ denote the activation function. θ represents the learnable hyperparameters of the network. Then, a physics-informed model can be trained by minimizing the following composite loss function:

$$\mathscr{L}(\theta)=\lambda_{pde}\mathscr{L}_{pde}(\theta)+\lambda_{ic}\mathscr{L}_{ic}(\theta)+\lambda_{bc}\mathscr{L}_{bc}(\theta),$$

$$\mathscr{L}_{pde}(\theta)=\frac{1}{N_r}\sum_{i=1}^{N_r}\left|\mathscr{T}\left[u_\theta\left(\boldsymbol{x}_r^i,t_r^i\right)\right]-f\left(\boldsymbol{x}_r^i,t_r^i\right)\right|^2,$$

$$\mathscr{L}_{bc}(\theta)=\frac{1}{N_{bc}}\sum_{i=1}^{N_{bc}}\left|\mathscr{B}\left[u_\theta\left(\boldsymbol{x}_{bc}^i,t_{bc}^i\right)\right]-g\left(\boldsymbol{x}_{bc}^i,t_{bc}^i\right)\right|^2,$$

$$\mathscr{L}_{ic}(\theta)=\frac{1}{N_{bc}}\sum_{i=1}^{N_{bc}}\left|\mathscr{H}\left[u_\theta\left(\boldsymbol{x}_{ic}^i,t_{ic}^i\right)\right]-h\left(\boldsymbol{x}_{ic}^i,t_{ic}^i\right)\right|^2,$$

where $\left\{\left\{\boldsymbol{x}_r^i,t_r^i\right\},\left\{\boldsymbol{x}_{bc}^i,t_{bc}^i\right\},\left\{\boldsymbol{x}_{ic}^i,t_{ic}^i\right\}\right\}$ are train points for the solving domains, the boundaries, and the initial values using a sampling algorithm including LHS (McKay et al., 2000), random sampling, Sobol sequences (Sobol, 1967), etc. Lu et al. demonstrated that the choice of sampling method for a given task affects the accuracy of subsequent predictions (Wu et al., 2023b). They proposed a residual-based adaptive resampling method to ensure that a certain probability of configuration points with significant errors is retained after each batch of iterations, thereby improving the accuracy of PINN. N_r,N_{bc},N_{ic} correspond to the number of the respective configuration points.$\lambda_{pde},\lambda_{bc},\lambda_{ic}$ represent weighting factors for different loss function components, thereby gauging the significance of the optimization process. These weighting factors can be either fixed manually or dynamically adjusted during network training. The conventional PINN framework is shown in Fig. 1.

The present study examines the propagation of ultrasound waves in a two-dimensional medium and derives $\mathscr{T}[\cdot]=\frac{\partial^2}{\partial t^2}-c^2\left(\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}\right)$, $\mathscr{H}[\cdot]=1,\frac{\partial}{\partial t}$ from Eq. (2). $\mathscr{B}[\cdot]=0,\boldsymbol{n}\cdot\nabla$ include absorbing and reflecting boundaries. Meanwhile, the arbitrary order derivatives concerning the independent variables required by these operators can be computed by the automatic differentiation layer of the neural network. It is noteworthy that, in contrast to the conventional finite element method, PINNs, as a meshless method, simulate acoustic wave propagation with free boundary conditions that must be integrated with the remaining operators in the form of boundary operators to construct an optimization function. However, this methodology may result in the disappearance of the reflection due to imbalanced loss weights.

2.3. Physics-informed Fourier neural network (PIFNN)

The wave propagation in inhomogeneous medium is a highly complex physical process, and its scattering process causes wave distortion and mixing, which is manifested as the superposition of high and low frequencies on different spatial and temporal scales (Görtz et al., 2023). However, the spectral bias indicates that neural networks tend to learn low-frequency features during training, with deficiencies in learning multi-frequency or high-frequency features. The spectral bias of PINNs is much more severe due to the additional physics constraints, resulting in difficulties for PINNs in accurately capturing reflection and scattering features in simulated wave propagation. To mitigate the pathology of spectral bias, this subsection introduces a stochastic Fourier feature mapping of the coordinates of the training points to enhance the capability of PINNs in learning high-frequency features.

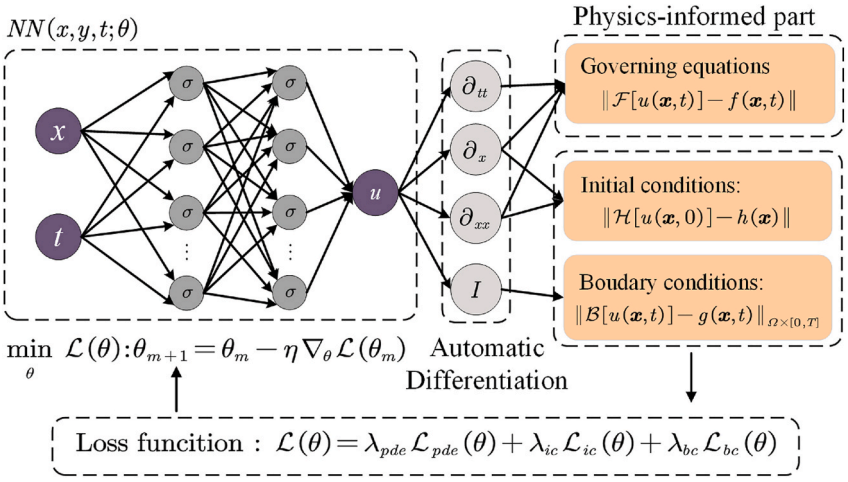


Fig. 1. Conventional PINN framework consisting of connected feedforward neural network and a physics-informed part.

The Random Fourier Feature (RFF) Mapping was initially conceived to approximate an arbitrary stationary kernel function, aiming to accelerate kernel machine training. Subsequently, Tancik et al. extended the method to deep neural networks by characterizing the input spatiotemporal coordinates to the hypersphere's surface (Tancik et al., 2020). The RFF mapping is denoted as:

$$\gamma_{(i)}(\boldsymbol{v}) = \begin{bmatrix} \cos(2\pi \boldsymbol{B}_{(i)} \boldsymbol{v}) \\ \sin(2\pi \boldsymbol{B}_{(i)} \boldsymbol{v}) \end{bmatrix}, i = 1, 2, \cdots, M,$$

where $\boldsymbol{B} = [b_1, b_2, \cdots b_m]^T \in \mathbb{R}^{m \times d}$, sampled from a Gaussian distribution $\mathbb{N}(0, \varepsilon^2)$, which implies that the probability of b taking a greater magnitude is directly proportional to the standard deviation ε of the distribution. Here, b represents the Fourier basis frequency vector. For the sine-cosine kernel function, a relatively large value of ε results in the emergence of high-frequency eigenfunctions, as well as the formation of narrow eigenvalue gaps. Consequently, Fourier features can effectively address the issue of spectral bias, facilitating a more rapid convergence towards the high-frequency components of the objective function (Tancik et al., 2020). The eigenvalues of the neural tangent kernel can be adjusted by varying the value of ε thereby controlling the frequency with which the neural network tends to learn. M is the number of RFF mapping,m means the number of neurons in the first hidden layer and d represent the dimension of the input spatio-temporal coordinates.

In the context of wave propagation problems with spatiotemporal features, the Fourier feature mappings are represented as a composition of temporal and spatial coordinates, which is

$$\gamma_{(i)}(\boldsymbol{x}) = \begin{bmatrix} \cos(2\pi \boldsymbol{B}_{(i)} \boldsymbol{x}) \\ \sin(2\pi \boldsymbol{B}_{(i)} \boldsymbol{x}) \end{bmatrix}, i = 1, 2, \cdots, M_x,$$

$$\gamma_{(j)}(t) = \begin{bmatrix} \cos(2\pi \boldsymbol{B}_{(j)} t) \\ \sin(2\pi \boldsymbol{B}_{(j)} t) \end{bmatrix}, j = 1, 2, \cdots, M_t,$$

where $\gamma_{(i)}(\boldsymbol{x})$ and $\gamma_{(j)}(t)$ stand for spatial and temporal Fourier feature mappings, respectively. Each of $\boldsymbol{B}_{(i)}$ and $\boldsymbol{B}_{(j)}$ sampled from $\mathbb{N}(0, \varepsilon_i^x)$ and $\mathbb{N}(0, \varepsilon_i^t)$,respectively. it's worth noting that $\boldsymbol{B}_{(i)}$ and $\boldsymbol{B}_{(j)}$ can be set by the user to be trainable as a hyperparameter or not. In this work, $\boldsymbol{B}_{(i)}$ and $\boldsymbol{B}_{(j)}$ remains constant after sampling, which means that the proposed Fourier features do not add any additional training cost.

Based on the above analysis, a novel multi-scale network architecture, named Physics-informed Fourier neural networks (PIFNN), was devised to approximate the solution to the PDE. In contrast to conventional PINNs, PIFNN incorporates a Fourier feature layer preceding the hidden layer of the neural network, thereby enhancing the capacity to learn high-frequency functions. Additionally, a spatiotemporal multi-scale feature layer was constructed following the hidden layer, facilitating the integration of Fourier features sampled in temporal and spatial coordinates from diverse ε. Specifically, the feed-forward pass of the network is defined as

$$\boldsymbol{H}_{\boldsymbol{x},i}^1 = \sigma\left(w^1 \cdot \gamma_i(\boldsymbol{x}) + b^1\right), i = 1, 2, \ldots, M_x,$$

$$\boldsymbol{H}_{t,j}^1 = \sigma\left(w^1 \cdot \gamma_j(t) + b^1\right), j = 1, 2, \ldots, M_t,$$

$$\boldsymbol{H}_{\boldsymbol{x},i}^l = \sigma\left(w^l \cdot \boldsymbol{H}_{\boldsymbol{x},i}^{l-1} + b^l\right), l = 2, \ldots, L \sim \text{and} \sim i = 1, 2, \ldots, M_x,$$

$$\boldsymbol{H}_{t,j}^l = \sigma\left(w^l \cdot \boldsymbol{H}_{t,j}^{l-1} + b^l\right), l = 2, \ldots, L \sim \text{and} \sim j = 1, 2, \ldots, M_t,$$

$$\boldsymbol{H}_{ij}^L = \boldsymbol{H}_{\boldsymbol{x},i}^L \odot \boldsymbol{H}_{t,j}^L, i = 1, 2, \ldots, M_x \sim \text{and} \sim j = 1, 2, \ldots, M_t,$$

$$u_\theta(\boldsymbol{x}, t) = w^{L+1} \cdot \left[\boldsymbol{H}_{1,1}^L, \ldots, \boldsymbol{H}_{M_x, M_t}^L\right] + b^{L+1},$$

where $\left\{w^l, b^l\right\}$ represent weights and bias of the l-th layer, initialized at the commencement of the training phase, with the weight initialization directly influencing the network convergence. An inappropriate initialization strategy may result in the network training gradient becoming fixed over time, leading to either gradient explosion or gradient vanishing. The Xavier initialization strategy is employed to mitigate the vanishing gradient phenomenon of the tanh function, thereby ensuring that the variance of the weights at each layer is maintained at a stable value (Glorot and Bengio, 2010). $\odot$ is the point-wise multiplication operator, the implementation of feature fusion. we emphasize that, for networks with Fourier features, ε determines the frequency that the networks prefer to learn. As a result, if we just employ networks with one Fourier feature embedding, then whatever the value of ε is used to initialize Fourier feature mappings, will yield slower convergence to the rest frequency components, except for the preferable frequencies determined by the choice of ε. Additionally, for time-independent PDEs such as the wave equation, multi-scale behavior may exist not only across spatial directions but also acrosstime. Therefore, it is reasonable to embed inputs to multiple temporal and spatial Fourier features with different ε and then concatenate them through a linear layer to ensure all frequency components can be learned with the same convergence rate.

This architectural design is inspired by the Fourier spectral method (Zhou et al., 2022), as shown in Fig. 2. To better comprehend the rationale behind the architecture, suppose that $u_\theta(\boldsymbol{x}, t)$ represents an approximation of $u(\boldsymbol{x}, t)$, which is the analytical solution to the PDE, whose Fourier decomposition is denoted as

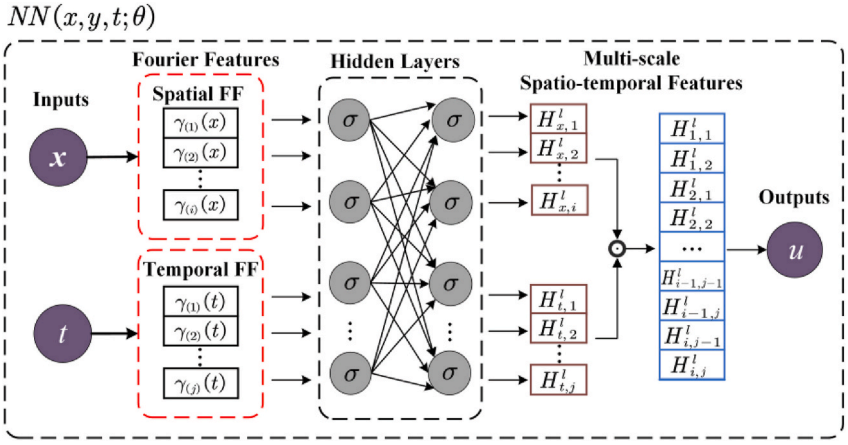


Fig. 2. Spatio-temporal multi-scale Fourier feature architecture.

$$u(\boldsymbol{x},t)=\sum_{k=-\infty}^{\infty}\widehat{u}_k(t)e^{ik\boldsymbol{x}},$$

where $\widehat{u}_k(t)$ is Fourier coefficient corresponding to the wave-number k at time t. Note that Eq. (19) shows that $u_\theta(\boldsymbol{x},t)$ is simply a linear combination of $\left\{\boldsymbol{H}^L_{1,1},\ldots,\boldsymbol{H}^L_{M_x,M_t}\right\}$, which can be rewritten in a consistent form as follows:

$$u_\theta(\boldsymbol{x},t)=\sum_{i=1}^{M_x}\sum_{j=1}^{M_t}w^{L+1}_{i,j}\boldsymbol{H}^L_{i,j}+b^{L+1}_{i,j}=\sum_{i=1}^{M_x}\sum_{j=1}^{M_t}w^{L+1}_{i,j}\boldsymbol{H}^L_{t,j}\odot\boldsymbol{H}^L_{\boldsymbol{x},i}+b^{L+1}_{i,j},$$

where $w^{L+1}_{i,j}$ and $b^{L+1}_{i,j}$ are the weights and bias of the L+1-th layer. In our framework, the spatial Fourier features $\boldsymbol{\gamma}_{(i)}(\boldsymbol{x})$ and temporal Fourier features $\boldsymbol{\gamma}_{(j)}(t)$ are independently processed by fully connected layers sharing the same weights, which project them into a unified spectral coefficient space. This design allows the network to learn how $u(\boldsymbol{x},t)$ can be expanded over spatial Fourier bases $e^{ik\boldsymbol{x}}$, while simultaneously estimating the corresponding time-varying coefficients $\widehat{u}_k(t)$. At the final muti-scale spatiotemporal features layer, these learned coefficients are explicitly fused, forming a coherent multiscale representation that mirrors the discrete Fourier series form. Consequently, the network approximation $u(\boldsymbol{x},t)$ from separable spatial and temporal Fourier features, enables interpretable learning of multiscale dynamics consistent with the Fourier spectral method.

2.4. Adaptive loss weight training algorithm for PINNs/PIFNNs

The Neural Tangent Kernel (NTK) theory elucidates that the eigenvalues of the NTK eigenfunctions in different directions determine the rate of convergence of the deep neural network along the corresponding direction (Jacot et al., 2018). Consequently, the NTK of each component of the PINNs composite loss function can be designed to achieve equilibrium between the respective convergence speeds. In PINN, considering the loss function described in Eq. (2) is minimized by gradient descent with an infinitesimally small learning rate. In accordance with the derivations presented by Wang et al. (2022), the training dynamics of PINNs can be expressed as the following ODE system:

$$\begin{bmatrix}\dfrac{d\mathscr{T}\left[u_\theta(\boldsymbol{x}^i_r,t^i_r)\right]}{dt}\\ \dfrac{d\mathscr{B}\left[u_\theta(\boldsymbol{x}^i_{bc},t^i_{bc})\right]}{dt}\\ \dfrac{d\mathscr{H}\left[u_\theta(\boldsymbol{x}^i_{ic},t^i_{ic})\right]}{dt}\end{bmatrix}=\boldsymbol{K}(t)\cdot\begin{bmatrix}\mathscr{T}\left[u_\theta(\boldsymbol{x}^i_r,t^i_r)\right]-f(\boldsymbol{x}^i_r,t^i_r)\\ \mathscr{B}\left[u_\theta(\boldsymbol{x}^i_{bc},t^i_{bc})\right]-g(\boldsymbol{x}^i_{bc},t^i_{bc})\\ \mathscr{H}\left[u_\theta(\boldsymbol{x}^i_{ic},t^i_{ic})\right]-h(\boldsymbol{x}^i_{ic},t^i_{ic})\end{bmatrix},$$

$$\boldsymbol{K}(t)=\begin{bmatrix}\boldsymbol{K}_{rr}(t) & \boldsymbol{K}_{ric}(t) & \boldsymbol{K}_{rbc}(t)\\ \boldsymbol{K}_{icr}(t) & \boldsymbol{K}_{icic}(t) & \boldsymbol{K}_{icbc}(t)\\ \boldsymbol{K}_{bcr}(t) & \boldsymbol{K}_{bcic}(t) & \boldsymbol{K}_{bcbc}(t)\end{bmatrix}=\begin{bmatrix}\dfrac{\lambda_r}{N_r}J_r(t)\\ \dfrac{\lambda_{bc}}{N_r}J_{bc}(t)\\ \dfrac{\lambda_r}{N_r}J_{ic}(t)\end{bmatrix}\begin{bmatrix}J^T_r(t) & J^T_{bc}(t) & J^T_{ic}(t)\end{bmatrix},$$

$$J_r(t)=\frac{\partial\mathscr{T}\left[u_\theta(\boldsymbol{x}^i_r,t^i_r)\right]}{\partial\theta},$$

$$J_{bc}(t)=\frac{\partial\mathscr{B}\left[u_\theta(\boldsymbol{x}^i_{bc},t^i_{bc})\right]}{\partial\theta},$$

$$J_{ic}(t)=\frac{\partial\mathscr{H}\left[u_\theta(\boldsymbol{x}^i_{ic},t^i_{ic})\right]}{\partial\theta},$$

$$(\boldsymbol{K}_{rr})_{ij}(t)=\langle\frac{\partial\mathscr{T}\left[u_\theta(\boldsymbol{x}^i_r,t^i_r)\right]}{\partial\theta},\frac{\partial\mathscr{T}\left[u_\theta(\boldsymbol{x}^j_r,t^j_r)\right]}{\partial\theta}\rangle,$$

$$(\boldsymbol{K}_{bcbc})_{ij}(t)=\langle\frac{\partial\mathscr{B}\left[u_\theta(\boldsymbol{x}^i_{bc},t^i_{bc})\right]}{\partial\theta},\frac{\partial\mathscr{B}\left[u_\theta\left(\boldsymbol{x}^j_{bc},t^j_{bc}\right)\right]}{\partial\theta}\rangle,$$

$$(\boldsymbol{K}_{icic})_{ij}(t)=\langle\frac{\partial\mathscr{H}\left[u_\theta(\boldsymbol{x}^i_{ic},t^i_{ic})\right]}{\partial\theta},\frac{\partial\mathscr{H}\left[u_\theta\left(\boldsymbol{x}^j_{ic},t^j_{ic}\right)\right]}{\partial\theta}\rangle,$$

where $J_r(t),J_{bc}(t),J_{ic}(t)$ stand for the Jacobian matrices of $\mathscr{T}[\cdot],\mathscr{B}[\cdot],\mathscr{H}[\cdot]$ with respect to θ respectively. $\boldsymbol{K}(t)$ is the NTK of PINN, which is a positive semi-definite matrix. Wang et al. demonstrated that the kernel $\boldsymbol{K}(0)$ converges probabilistically to a deterministic kernel matrix as the network width tends to infinity (Wang et al., 2022). Moreover, The overall convergence rate of the composite loss function is determined by $\boldsymbol{K}(t)$, while $\mathscr{T}\left[u_\theta(\boldsymbol{x}^i_r,t^i_r)\right],\mathscr{B}\left[u_\theta(\boldsymbol{x}^i_{bc},t^i_{bc})\right]$ and $\mathscr{H}\left[u_\theta(\boldsymbol{x}^i_{ic},t^i_{ic})\right]$ are determined by $\boldsymbol{K}_{rr}(t),\boldsymbol{K}_{bcbc}(t)$ and $\boldsymbol{K}_{icic}(t)$ respectively. Accordingly, it is feasible to adaptively adjust the loss component gradient flow by assessing the loss term's significance through NTK calculation. Further details on the NTK derivation and theoretical background can be found in Ref. (Wang et al., 2022).

The adaptive loss weights can be computed by singular value decomposition in the form of

$$\lambda_r=\frac{Tr(\boldsymbol{K}_{rr})+Tr(\boldsymbol{K}_{bcbc})+Tr(\boldsymbol{K}_{icic})}{Tr(\boldsymbol{K}_{rr})},$$

$$\lambda_{bc} = \frac{Tr(\boldsymbol{K}_{rr}) + Tr(\boldsymbol{K}_{bcbc}) + Tr(\boldsymbol{K}_{icic})}{Tr(\boldsymbol{K}_{bc})},$$

$$\lambda_{ic} = \frac{Tr(\boldsymbol{K}_{rr}) + Tr(\boldsymbol{K}_{bcbc}) + Tr(\boldsymbol{K}_{icic})}{Tr(\boldsymbol{K}_{ic})},$$

where $Tr(\cdot)$ represents the trace of the matrix.

Algorithm 1. Adaptive loss weights for PINNs/PIFNNs

Step1: Consider a fully connected neural network to approximate latent physical field $u_\theta(\boldsymbol{x}, t)$ with hyperparameters θ

Step2: Define an initial margin problem and compute the loss function by automatic differentiation:

$$\mathscr{L}(\theta) = \lambda_{pde}\mathscr{L}_{pde}(\theta) + \lambda_{ic}\mathscr{L}_{ic}(\theta) + \lambda_{bc}\mathscr{L}_{bc}(\theta),$$

Step3: Initialization of hyperparameters and Fourier feature mappings, Configuration points for the number of neural tangent kernels sampled from the computational domain $\{\{\boldsymbol{x}_r^i, t_r^i\}, \{\boldsymbol{x}_{bc}^i, t_{bc}^i\}, \{\boldsymbol{x}_{ic}^i, t_{ic}^i\}\}$.

Step4: Use m gradient descent iterations to update the hyperparameters θ and the loss weights $\lambda_{pde}, \lambda_{bc}$ and λ_{ic} as:

for m = 1 to M **do**

(a) Compute $\lambda_{pde}, \lambda_{bc}$ andλ_{ic} by Eq. (29), Eq. (30) and Eq. (31)

$$\lambda_{pde}(m) = \frac{Tr(\boldsymbol{K}_{rr}(m)) + Tr(\boldsymbol{K}_{bcbc}(m)) + Tr(\boldsymbol{K}_{icic}(m))}{Tr(\boldsymbol{K}_{rr}(m))}$$

$$\lambda_{bc}(m) = \frac{Tr(\boldsymbol{K}_{rr}(m)) + Tr(\boldsymbol{K}_{bcbc}(m)) + Tr(\boldsymbol{K}_{icic}(m))}{Tr(\boldsymbol{K}_{bc}(m))}$$

$$\lambda_{ic}(m) = \frac{Tr(\boldsymbol{K}_{rr}(m)) + Tr(\boldsymbol{K}_{bcbc}(m)) + Tr(\boldsymbol{K}_{icic}(m))}{Tr(\boldsymbol{K}_{ic}(m))}$$

Where $\boldsymbol{K}_{rr}(t), \boldsymbol{K}_{bcbc}(t)$ and $\boldsymbol{K}_{icic}(t)$ calculate by Eq. (29), Eq. (30) and Eq. (31)

(b) Update the hyperparameters θ via Adam optimizer

$\theta_{m+1} = \theta_m - \eta\nabla_\theta\mathscr{L}(\theta_m)$ **end**

3. Results and discussion

In this section, the proposed method is validated through numerical experiments; exploring the method on a simple string vibration problem is a prerequisite for extending it to complex problems. Then, the proposed method is evaluated in a two-dimensional inhomogeneous medium to assess its predictive ability for ultrasonic propagation. In the tests, the numerical solutions obtained by FDTD are considered as the reference solutions to evaluate the effectiveness and accuracy of the prediction results of PIFNN. The proposed PIFNN is compared with the traditional PINN methods. To fairly compare the performance of the different models, the same hidden layer parameters and training points are used for PIFNN and PINN. The neural network approximates the solution of the PDE by minimizing a composite loss function to predict the wavefield at any time and location, requiring two early-time snapshots as training data. Undeniably, the training data for the initial conditions set by the network are derived from sampling the wavefield data obtained from analytical formulations or finite differences, which are intended to constitute a complete initial marginal problem. It is worth noting that other data-driven methods may not accurately predict ultrasonic propagation with wavefields using only two early time snapshots, demonstrating the superiority of the PIFNN method for solving the wave equation with a small amount of data. Due to the non-convex nature of the optimization problem, it is not possible to find a unique solution. It follows that the neural network's hyperparameters can impact the training results, and the hyperparameters for all experiments were obtained after tuning. All numerical experiments were conducted with the Adam (Kingma and Ba, 2014) during optimization were coded in Pytorch 1.13 (Paszke et al., 2019) with Python 3.9. These numerical experiments were performed on a computer with an NVIDIA RTX A6000 card (48 GB) and an Intel 4210R central processing unit (CPU). It is worth noting that Appendix A provides the detailed configuration of the networks used in the experiment. The accuracy of the predictions is measured by the relative L^2 error.

$$L^2 error = \frac{\sqrt{\sum_{i=1}^{N} |u_\theta(\boldsymbol{x}_i, t_i) - u(x_i, t_i)|^2}}{\sqrt{\sum_{i=1}^{N} |u(x_i, t_i)|^2}}.$$

3.1. One-dimensional wave propagation

This subsection elucidates the training pathology of the conventional PINNs on multiscale methodology for wave problems and corroborates the precision of the proposed PIFNN on simple string vibration problem. This problem is described by the one-dimensional form of Eq. (2). Specifically, the PDE can be expressed as

$$u_{tt}(x,t) - 4u_{xx}(x,t) = 0, x \in [0,1], t \in [0,1].$$

Boundary conditions:

$$u(0,t) = u(1,t) = 0, t \in [0,1].$$

Initial conditions:

$$u(x,0) = \sin(\pi x) + \sin(4\pi x), x \in [0,1],$$

$$u_t(x,0) = 0, x \in [0,1].$$

The solution to the problem, derived through the method of separated variables, is provided below:

$$u(x,t) = \sin(\pi x)\cos(2\pi t) + \sin(4\pi x)\cos(8\pi t).$$

In particular, the initial conditions Eq. (36) serve as special boundary conditions on the spatiotemporal domain Ω, as they are both restricted to terms of the same order as u.

$$u(x_u, t_u) = g(x_u, t_u), x_u, t_u \in \partial\Omega,$$

As demonstrated by Eq. (7), the total loss function of the problem can be expressed as follows:

$$\mathscr{L}(\theta) = \lambda_{pde}\frac{1}{N_r}\sum_{i=1}^{N_r}\left|\frac{\partial^2 u_\theta(\boldsymbol{x}_r^i, t_r^i)}{\partial t^2} - 4\frac{\partial^2 u_\theta(\boldsymbol{x}_r^i, t_r^i)}{\partial x^2}\right|^2 + \lambda_{bc}\frac{1}{N_u}\sum_{i=1}^{N_u}|u_\theta(\boldsymbol{x}_u^i, t_u^i) - g(\boldsymbol{x}_u^i, t_u^i)|^2 + \lambda_{ic}\frac{1}{N_{u_t}}\sum_{i=1}^{N_{u_t}}\left|u_\theta\left(\boldsymbol{x}_{u_t}^i, t_{u_t}^i\right)\right|^2,$$

Where the number of training points for each part of the loss function is set to $N_r = N_u = N_{u_t} = 240$ and all training points $\left\{(\boldsymbol{x}_u^i, t_u^i), (\boldsymbol{x}_r^i, t_r^i), \left(\boldsymbol{x}_{u_t}^i, t_{u_t}^i\right)\right\}$ are uniformly sampled from the corresponding computational domain. $\lambda_{pde}, \lambda_{bc}$ andλ_{ic} are updated every 100 iterations by calculating NTK of networks according to Algorithm 1.

The three network frameworks, including conventional PINN, PINN with adaptive activation function (aPINN) (Jagtap et al., 2020), and Physics-informed Fourier neural networks with adaptive loss weight (Alw-PIFNN), with the same parameters of three layers of 200 neurons, and their predictions were compared with the exact solution after 40, 000 iterations of the Adam optimizer with a learning rate of 1e-3, as shown in Fig. 3. For a Fourier feature network, a spatial Fourier feature mapping is initialized by $\varepsilon_1^x = 1$ and two temporal Fourier feature mappings are initialized by $\varepsilon_1^t = 1, \varepsilon_2^t = 4$. As shown in Fig. 3, the conventional PINN is capable of predicting only the low-frequency components of wave propagation, resulting in inaccurate representations of the overall wave dynamics. The aPINN enhances the model's ability to learn high-frequency components through its adaptive activation, yet its predictions still deviate from the exact solution. In contrast, the proposed PFINN achieves accurate prediction of wave propagation across both low and high frequency components, demonstrating superior fidelity and generalization capability.

Fig. 4 illustrates that the disparate loss terms of PINN and PIFNN

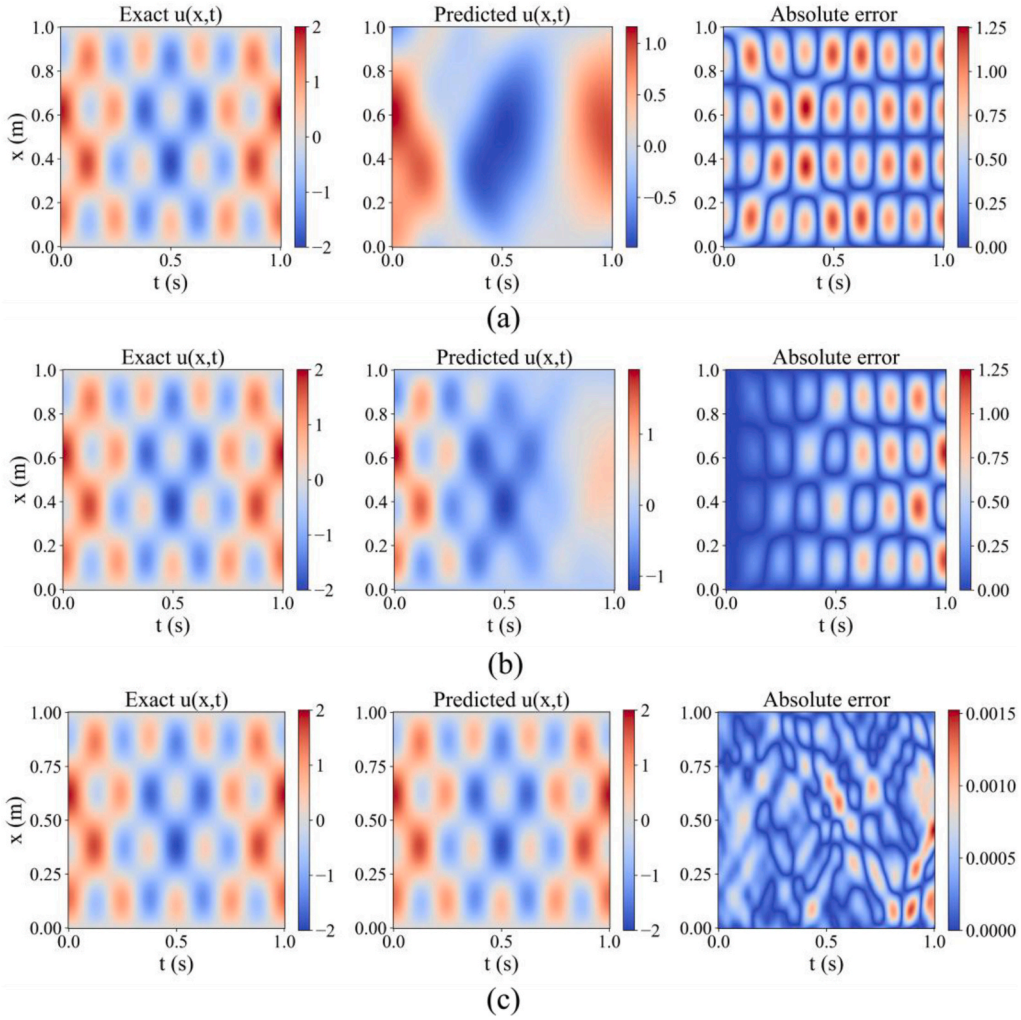


Fig. 3. 1D wave equation:(a) Comparison of the PINN prediction's results to the exact solution. (b) Comparison of aPINN 's prediction results to the exact solution. (c) Comparison of the PFINN's prediction results to the exact solution.

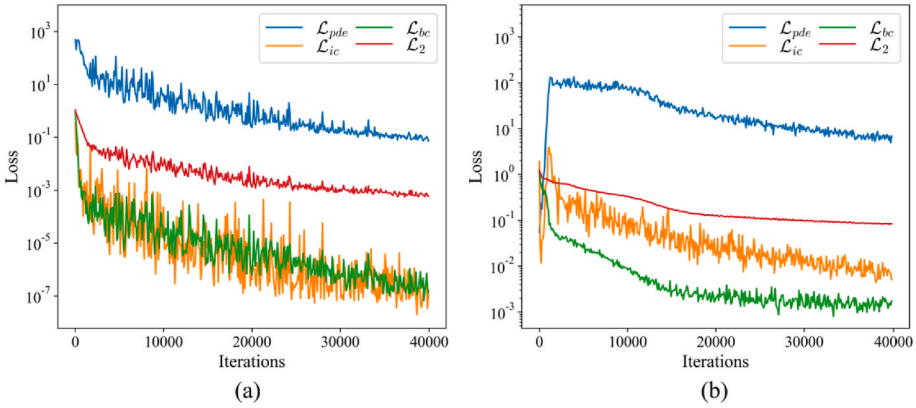


Fig. 4. 1D wave equation: (a) Convergence of the different terms in the loss function of PIFNN and the relative L^2 error during training. (b) Convergence of the different terms in the loss function of PINN and the relativeL^2 error during training.

utilizing the adaptive weighting strategy exhibit a comparable convergence rate, indicating that the pathology resulting from the imbalance in loss convergence rates, as previously observed by Wang et al. (2022), has been mitigated. It may be hypothesized that the reason why PINN still fails may be due to the limitation of spectral bias inherent in fully connected networks. In contrast, the success of PIFNN is predicted to be attributed to the enhanced capacity to learn high-frequency functions due to the appropriate spatiotemporal Fourier feature mapping. The aforementioned hypothesis is further substantiated by Fig. 5, which depicts the eigenvalues K_r, K_{bc} and K_{ic} of the PIFNN following distinct iterations of the Adam optimizer. It can be observed that the order of magnitude of K_r is considerably larger than that of K_{bc} and K_{ic}, indicating

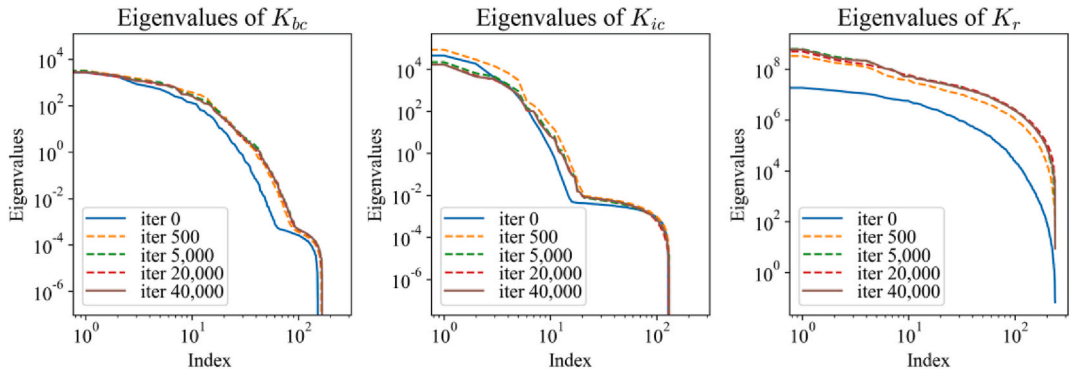


Fig. 5. Eigenvalues of K_{bc}, K_{ic} and K_r at different iterations during PIFNN training.

that the unadjusted convergence rate of the partial differential equation is more rapid than the rate of convergence of the initial and boundary conditions. This may impede the network's ability to converge to the desired solution. Nevertheless, the implementation of Algorithm 1 allows for the modification of the eigenvalues of the neural tangent kernel to a comparable magnitude, thereby facilitating a balanced rate of convergence. The dynamic adjustment of the weights is demonstrated in Fig. 6. Another observation of Fig. 5 is that K_r, K_{bc} and K_{ic} are not stationary, and show a tendency to move 'outward' or 'inward' as the training iterations progress. This suggests that the neural tangents have converged to a defined kernel with minimal alterations during the training phase. Such a suggestion is further confirmed by Fig. 6, which shows that the later stages of the loss weight evolution remain largely static over the ten independent repetitions of the Xavier weight initialization. Furthermore, the consistency of the results suggests that the adaptive weighting strategy is not sensitive to initialization.

The initialization of the Fourier feature mapping is of great importance, as it determines the frequency with which the network learns. Further analysis was conducted to investigate the impact of different initializations of the Fourier feature function on the prediction performance, considering 10 independent trials with different initializations for $\varepsilon_2^t \epsilon [0.1, 50]$ and reporting the L^2 error relative to the reference solution in Eq. (38) in Fig. 7. It can be observed that the Fourier feature mapping initialized by ε_2^t that is either excessively large or insufficiently small can have a deleterious impact on the network. Consequently, it is essential to select a suitable value of ε^x and ε^t to guarantee that the

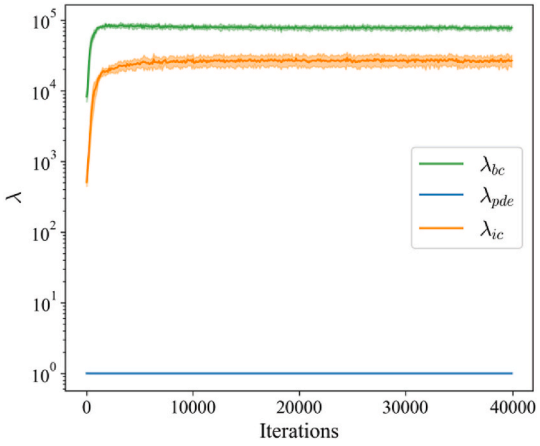


Fig. 6. The evolution of λ_{bc}, λ_{pde} and λ_{ic} obtained from ten independently repeated trials, updated by each of the 100 iterations. The green area and the yellow area represent the range of one standard deviation of λ_{bc} and λ_{ic}, respectively, statistically derived from ten trials. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

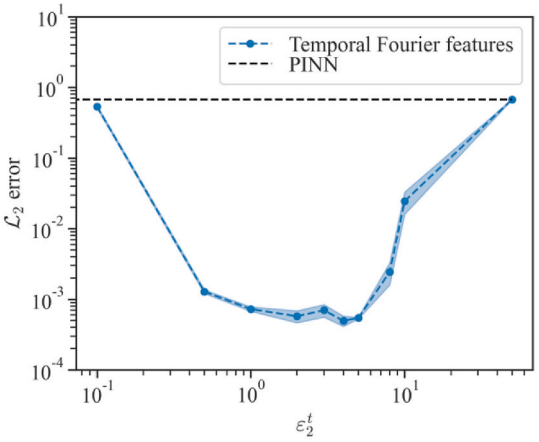


Fig. 7. 1D wave equation: Relative L_2 error of prediction for ten independent trials with different initialization for $\varepsilon_2^t \epsilon [0.1, 50]$.The black dashed line indicates the prediction relative L_2 error of PINN as the baseline.

PIFNN demonstrates optimal performance and accuracy in solving the wave equation in terms of a high or multiple frequency. In addition, ablation experiments were performed on the Fourier feature mapping and adaptive weighting strategies, and the relative L^2 errors of the prediction results of the four network frameworks are shown in Table 1.

3.2. Two-dimensional wave propagation

This subsection examines the PIFNN in modeling the propagation of ultrasound waves in a horizontally layered medium with three distinct velocities. During propagation, ultrasound waves may undergo expansion or compression due to changes in velocity, as well as reflection and transmission at interfaces. A comparison of the ability to capture the wavefield's complex dynamics, between PIFNN and conventional PINNs is presented to demonstrate the proposed training strategy's efficacy. Furthermore, the impact of varying the weights of the PIFNNs on the prediction accuracy is investigated, and the training dynamics of the PIFNN are elucidated through an eigenvalue analysis of the NTK. This subsection focuses on wave propagation in horizontally layered medium described by the two-dimensional (2D) form of Eq. (2). Specifically, the PDE is expressed as

$$\frac{\partial^2 u}{\partial t^2} - c(x,y)^2 \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) = 0,$$

where $c(x,y)$ is velocity field of 2D medium is shown in Fig. 8.

Fig. 8 illustrates the setup of the velocity model and transducer positions for three layers of acoustic medium with sound velocities of 1400 m/s,1200 m/s, and 1000 m/s, respectively, in sequence. The point

Table 1
Ablation experiments with the mean and standard deviation of the L^2 relative error of the wavefield predicted by different network architectures. The L^2 relative error of the optimal performance is bolded.

Model	PINN	Alw-PINN	PIFNN	Alw-PIFNN	aPINN
1D model	67.35 ± 1.73	6.63 ± 0.78	0.06 ± 0.01	**0.05 ± 0.01**	44.79 ± 1.21
Layered model	8.42 ± 1.24	5.53 ± 0.97	2.81 ± 0.32	**2.11 ± 0.21**	6.84 ± 0.51

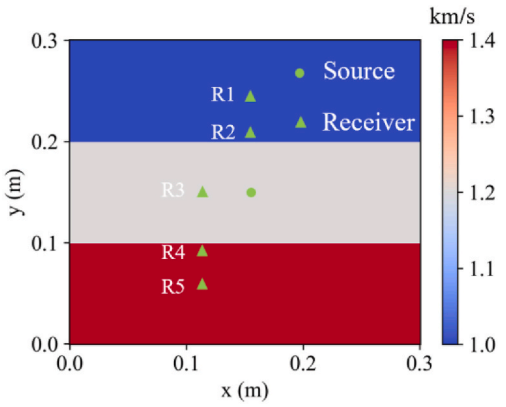


Fig. 8. Horizontally layered model, as well as the market of the location of the excitation source and the sensor. the receivers are located at R1 (0.15m, 0.24 m), R2 (0.15 m, 0.21m), R3 (0.12 m, 0.15 m), R4 (0.12 m, 0.09 m), R5(0.12 m, 0.06 m).

source of the Ricker wavelet time function with a main frequency of 30 kHz is located in the center of the region for excitation. In this case, the absorption conditions were set on all boundaries. It is noteworthy that when no additional boundary condition constraints are set, the PIFNN automatically satisfies the absorption boundaries due to the continuous smoothness and infinite differentiability of its solutions. As demonstrated by Eq. (7), the compote loss function of the case can be expressed as follows:

$$\begin{aligned}\mathscr{L}(\theta) = \lambda_{pde}\frac{1}{N_r}\sum_{i=1}^{N_r}\left|\frac{\partial^2 u_\theta(x_r^i, y_r^i, t_r^i)}{\partial t^2} - c^2(x_r^i, y_r^i)\Delta u_\theta(x_r^i, y_r^i, t_r^i)\right|^2 \\ + \lambda_{ic1}\frac{1}{N_{ic1}}\sum_{i=1}^{N_{ic1}}\left|u_\theta(x_{ic1}^i, y_{ic1}^i, t_{ic1}^i) - u(x_{ic1}^i, y_{ic1}^i, t_{ic1}^i)\right|^2 \\ + \lambda_{ic2}\frac{1}{N_{ic2}}\sum_{i=1}^{N_{ic2}}\left|u_\theta(x_{ic2}^i, y_{ic2}^i, t_{ic2}^i) - u(x_{ic2}^i, y_{ic2}^i, t_{ic2}^i)\right|^2,\end{aligned}$$

where the training points for the PDE loss term are sampled in the spatiotemporal domain using the Sobol sequences. Two snapshots at t = 0.5s and t = 0.75s were used as initial conditions, where the number of configuration points sampled from each early snapshot data $N_{ic1} = N_{ic2} = 6400$. In order to achieve rapid convergence of the neural network, it is necessary to scale both the input wavefield snapshot data and the PDE loss function term(given in Appendix B). Considering the increase in dimensionality and complexity of the speed model, the hidden layer was adjusted to 6 layers of 100 neurons. Furthermore, the training points involved in the NTK computation were randomly sampled within the computational domain, and the number of samples was set to 200.

The four network frameworks, including conventional PINN, aPINN, PIFNN, and Physics-informed Fourier neural networks with adaptive loss weight (Alw-PIFNN), were trained with the Adam optimizer and a learning rate of 5e-4 over 20000 iterations. For PIFNN and Alw-PIFNN, the spatial and temporal Fourier feature mappings are initialized by $\varepsilon_1^t = \varepsilon_1^x = 1$. In order to conduct a fair comparison of the capabilities of PINN and PIFNN, the weights of the loss functions in Eq. (41) are set to $\lambda_{ic1} = \lambda_{ic2} = 10,000$ and $\lambda_{pde} = 1$. The 1.4ms ultrasonic propagation was simulated with a time step of 5e-7. their wavefield predictions were compared with the reference solution obtained from FDTD at 0.75ms, 1ms,1.25ms, and 1.4ms. as shown in Fig. 9. It is imperative that the reflected and transmitted waves at the interface are accurately captured in order to achieve refined inversion imaging. It can be observed that PINN does not accurately capture the reflected waves, whereas PIFNN is able to predict the reflected waves satisfactorily over a longer period of time, in comparison to the early snapshots. The analysis of the discrepancy between the predicted wavefield and the reference solution indicates that Alw-PIFNN attains the most accurate predictions, with a maximum error of approximately 3 %, as illustrated in Fig. 10.

Fig. 11 compares the training loss of the PINN and alw-PIFNN models, demonstrating that the alw-PIFNN loss function converges at a faster rate. The Fourier feature mapping and adaptive loss strategies facilitate more precise feature capture and faster convergence, which is crucial for modelling ultrasonic propagation in complex medium. This is further demonstrated in Fig. 12, which shows that time-domain signals received by the five transducers (R1–R5) shown in Fig. 8, during the ultrasonic propagation process predicted by Alw-PIFNN are almost identical to the reference solution. Additionally, Fig. 13 compares the frequency-domain characteristics of the R3 receiver signal predicted by different models. Among them, Alw-PIFNN demonstrates superior convergence behavior and accurately captures all frequency components, highlighting its advantage in mitigating spectral bias compared with other models.

To further analyze the robustness and effectiveness of the adaptive loss strategy, ten independent repetitive experiments were set up for different A to statistic the L2 error results obtained from Fig. 14. It is worth noting that the loss weights play a pivotal role in PINNs. Imbalances in loss weights can result in training failures, necessitating substantial experimental costs. In contrast, the adaptive strategy not only offers automatic tuning through analyzing the training dynamics of the network but also addresses one of the training pathologies of PINNs. To further clarify the training dynamics of PINNs, Fig. 15 illustrates the predicted wavefields of Alw-PIFNN at t = 1.4 ms for varying numbers of iterations. Alw-PIFNN captures the transmitted wave at the 3000 iteration, and as the number of iterations increases, the interlayer-induced reflected wave is better captured at the 7000 iteration. As evidenced by the train loss in Fig. 11(b), the loss shows a step at the corresponding calendar element, indicating convergence to a smaller solution space.

Fig. 16 shows the eigenvalues K_{ic1}, K_{ic2} andK_{pde} of PIFNN after different iterations of Adam's optimizer, It can be seen thatK_{pde} is orders of magnitude larger than K_{ic1} andK_{ic2}, suggesting that the unadjusted rate of convergence of the partial differential equation is faster than that of the initial conditions. Due to the same form of the loss function, K_{ic1} and K_{ic2} have approximately the same evolution pattern. Nevertheless, Algorithm 1 balances the convergence rates of different loss function terms. In addition, the neural tangent kernel is not fixed, unstable in the pre-training period, and the change gradually slows down with training iterations until it converges to a definite kernel, Correspondence in specific manifestation, which is the gradual convergence from the blue solid line at the begin of training to the brown solid line at the end of training, and the basic convergence at 15000 iterations. This manifestation is almost confirmed by the dynamic adjustment of the weights shown in Fig. 17, which shows that the evolution of the loss weights remains essentially unchanged at 15,000 iterations over ten independent iterations. In addition, ablation experiments were performed on the Fourier feature mapping and adaptive weighting strategies, and the

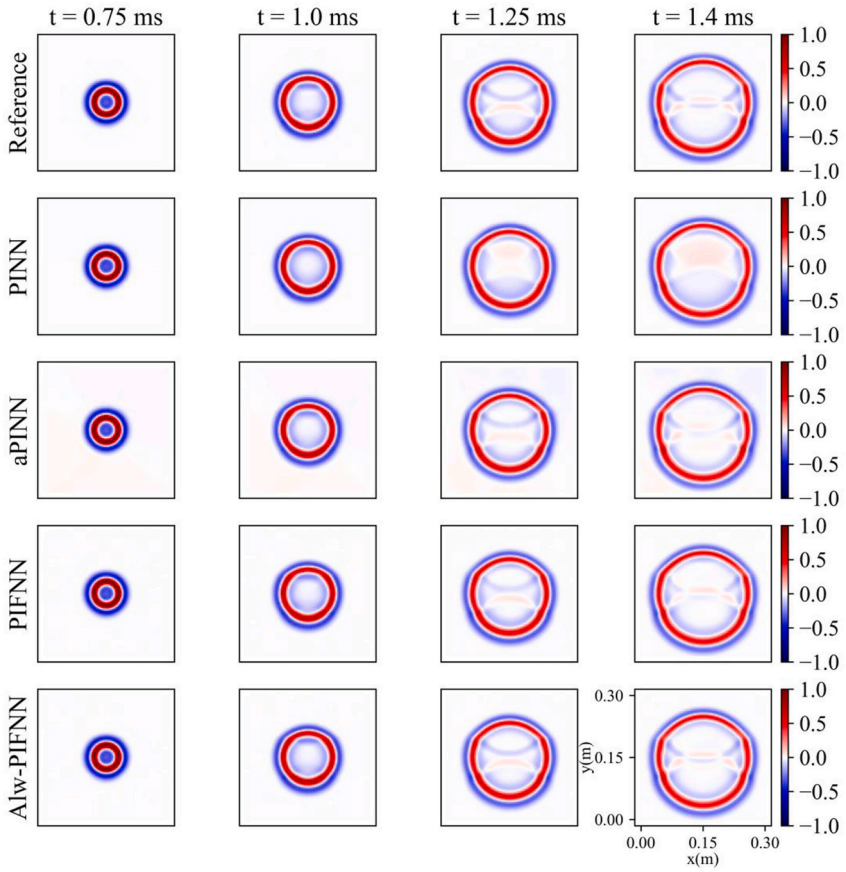


Fig. 9. Wavefield comparisons between FDTD, PINN, aPINN, PIFNN, and PIFNN using the adaptive loss weighting strategy(Alw-PIFNN) in the horizontally Layered velocity model.

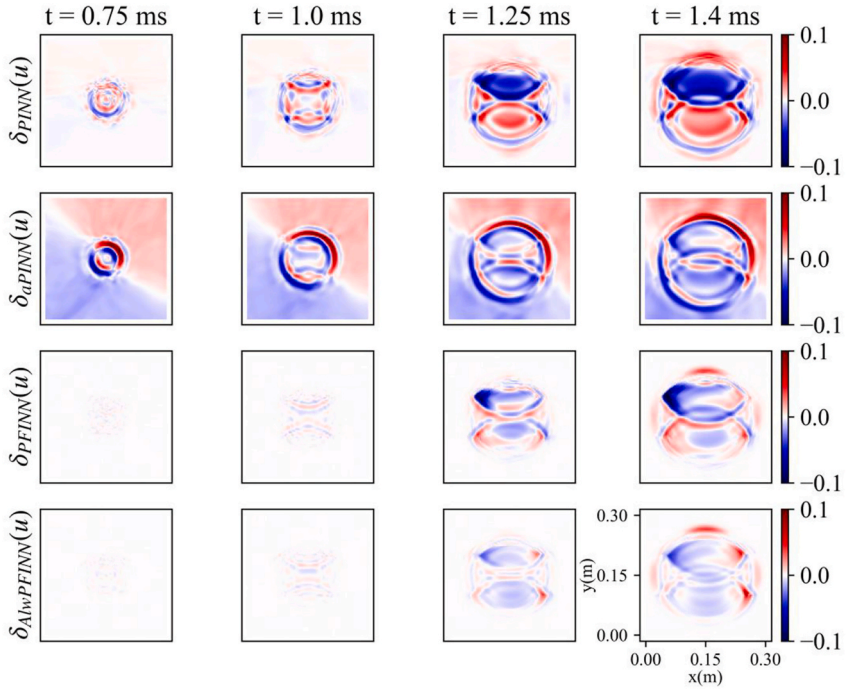


Fig. 10. The predicted wavefield errors of PINN, aPINN, PIFNN, and Alw-PIFNN compared to the reference solution.

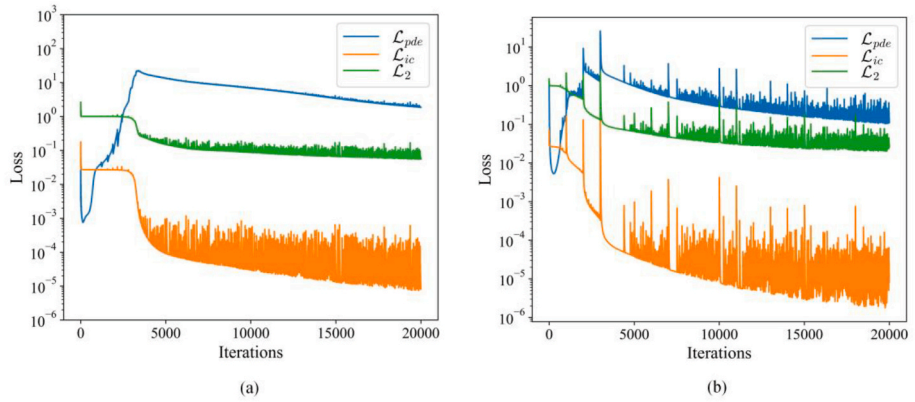


Fig. 11. (a) Convergence of the different terms in the loss function of PINN and the relative L^2 error during training. (b) Convergence of the different terms in the loss function of Alw-PIFNN and the relativeL^2 error during training.

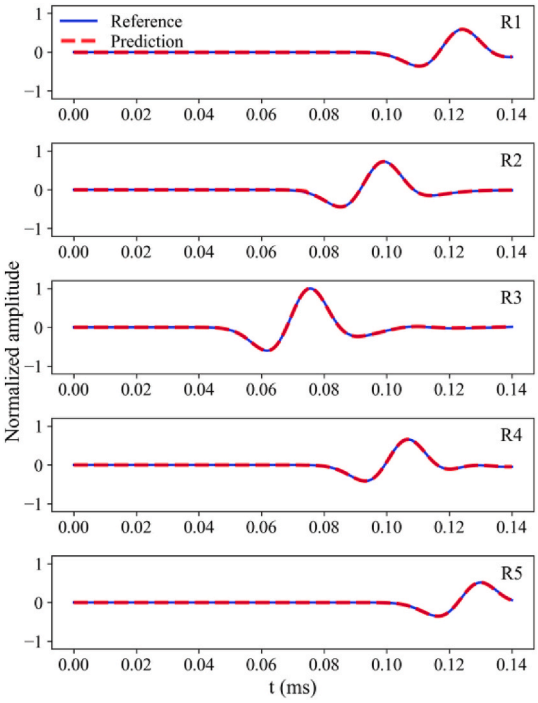


Fig. 12. Comparison of signals received by each receiver between Reference and Alw-PIFNN in horizontally layered velocity model.

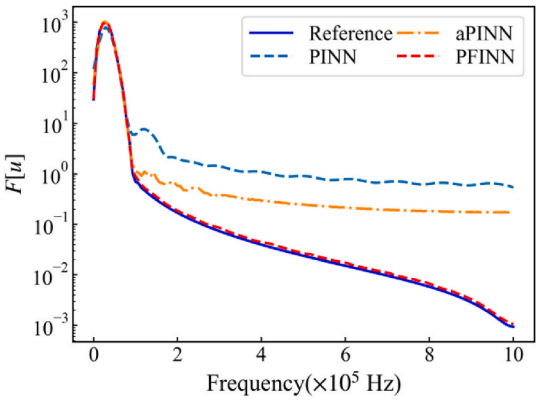


Fig. 13. Comparison of signals received by R3 receiver in frequency domain between Reference, PINN, aPINN, and Alw-PIFNN.

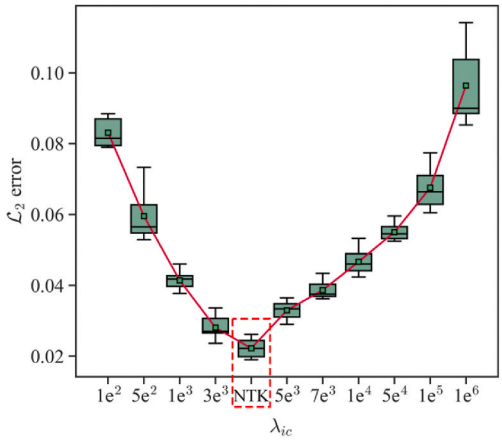


Fig. 14. The box diagram of the average relative L2 errors for four snapshots after 20000 Adam iterations, with the error due to different λ_{ic}. "NTK" boxed in red indicates adaptive adjustment weights of PIFNN using Algorithm 1. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

relative L2 errors of the wavefields prediction of the four network frameworks are shown in Table 1.

In discussing the limitations of the current method, we note that extending PIFNN to higher-dimensional or nonlinear wave propagation problems introduces additional computational and optimization challenges. Nevertheless, the Fourier feature mapping and NTK-based adaptive loss weighting increase training time by only ~0.6 % and parameters by ~200 kB (Table 2), indicating good scalability. Built upon PINNs, which are inherently designed for high-dimensional PDEs, PIFNN retains these advantages. Moreover, training strategies such as adaptive sampling, sequential training, and domain decomposition can be seamlessly integrated with PIFNN, further enhancing its potential applicability to high-dimensional and nonlinear problems.

4. Conclusion

In conclusion, this paper presents a novel architectural framework, PIFNN, designed to address the challenge of wave propagation. It integrates the Fourier feature layer and the adaptive loss weight method based on the NTK theory, thereby addressing the training pathologies caused by the dynamics of spectral differences and imbalance loss gradients.

Firstly, the training pathology of PINN is revealed in the one-dimensional wave equation when the objective function is multi-

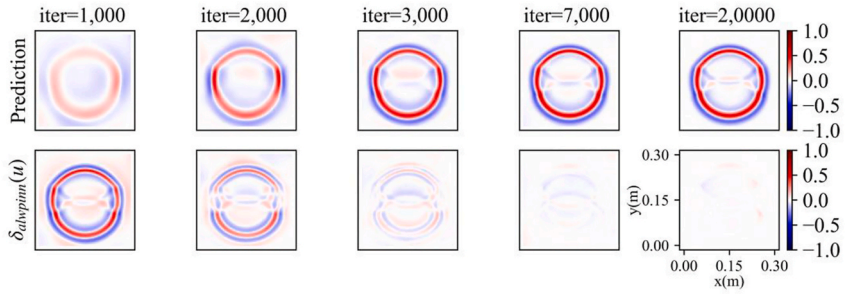


Fig. 15. Comparison of the PIFNN's wavefield predictions to the reference solution at different iterations for the Horizontally layered model.

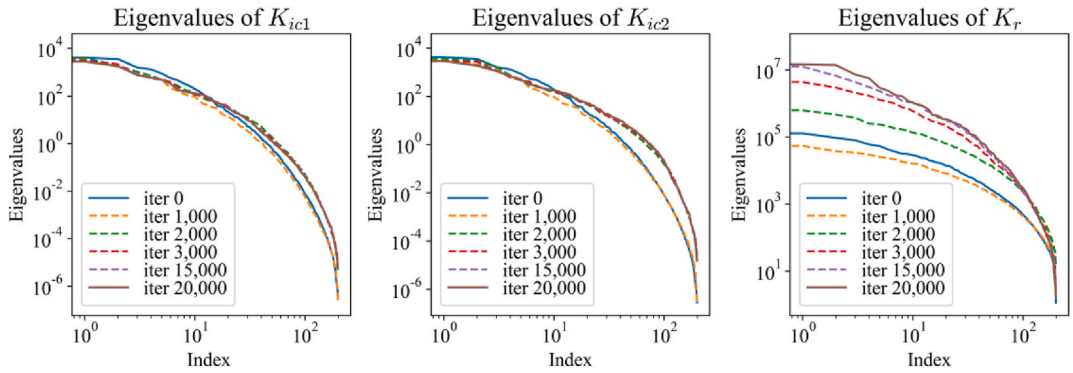


Fig. 16. Eigenvalues of K_{ic1}, K_{ic2} andK_r at different iterations during PIFNN training.

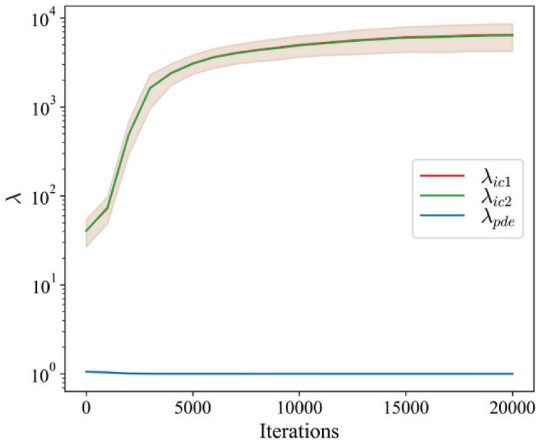


Fig. 17. The evolution of λ_{ic1}, λ_{ic2} and λ_{pde} obtained from ten independently repeated trials, updated by each of the 100 iterations. The green area and red area represent the range of one standard deviation of λ_{ic1} and λ_{ic2}, respectively, statistically derived from ten trials. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

frequency or high-frequency. After that, the proposed framework is used to accurately predict the multi-frequency string vibration problem, reaching a maximum error of approximately 2 %. Furthermore, the effects of different initialized Fourier eigenfunctions on the prediction performance are analyzed, demonstrating that the Fourier eigenfunctions are effective in enhancing the ability of PINN to learn the frequency-specific functions. Subsequently, the proposed methodology is extended to simulate the propagation of ultrasonic waves in a two-dimensional horizontal layered medium. Predictions of the wavefield at arbitrary times and locations are achieved using only two early-time snapshots. A comparison of the proposed method with PINNs demonstrates that the proposed method is capable of capturing reflection features with greater accuracy without additional training costs. The maximum error in the wavefield is approximately 3 %, which is an important prerequisite for inverse imaging. The gradient dynamics of the PIFNN are analyzed, and the effect of unpaired loss weights on the prediction performance is discussed, demonstrating the effectiveness and robustness of the neural cut kernel-based adaptive loss weighting strategy. The relationship between the training dynamics of the PIFNN and the accuracy of predicting ultrasonic propagation features is investigated. It is found that the PIFNN preferentially predicts transmitted waves and subsequently converges to the solution space with reflected waves.

Although the proposed method exhibits excellent performance in ultrasonic propagation, several challenges remain for broader deployment. The initialization of Fourier feature mappings and the extension to nonlinear or high-dimensional wave equations require further investigation to ensure robustness and scalability. Nevertheless, the core mechanisms, including Fourier feature enhancement and self-adaptive loss weighting, are inherently general and can be effectively transferred to other wave-based forward and inverse problems, including non-destructive testing, medical ultrasonic imaging, seismic inversion, and electromagnetic simulation, indicating the potential universality and cross-domain adaptability of the proposed approach for intelligent modeling of complex physical systems.

CRediT authorship contribution statement

Jiang Lu: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation. **Lishuai Liu:** Writing – review & editing, Supervision, Resources, Funding acquisition, Conceptualization. **Haiming Xu:** Software, Investigation, Conceptualization. **Chaoyu Sun:** Visualization, Data curation. **Yanxun Xiang:** Supervision, Project administration, Funding acquisition. **Fu-Zhen Xuan:** Investigation, Conceptualization.

Table 2
Training time and parameters for different frameworks in 2D wave propagation.

Model	PINN	Alw-PINN	PIFNN	Alw-PIFNN
Training time/iteration	0.5354s	0.5366s	0.5373s	0.5388s
parameters	202 KB	400 KB	205 KB	403 KB

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A

Table A1
Hyperparameters of PINNs/PIFNNs used for modeling wave propagation in this study.

	Hidden layers [width] × depth	Collocation points	Neural tangent kernel size	Loss weight parameters
1D model:PINN	[200] × 3	$N_r = N_u = N_{u_t} = 240$		$\lambda_{pde} = 1, \lambda_{bc} = 1e^5, \lambda_{ic} = 1e^4$
1D model:PINN	[200] × 3	$N_r = N_u = N_{u_t} = 240$	240	Adaptive loss weights
1D model:aPINN	[200] × 3	$N_r = N_u = N_{u_t} = 240$	n = 10	$\lambda_{pde} = 1, \lambda_{bc} = 1e^5, \lambda_{ic} = 1e^4$
1D model:PIFNN	[200] × 3	$N_r = N_u = N_{u_t} = 240$		$\lambda_{pde} = 1, \lambda_{bc} = 1e^5, \lambda_{ic} = 1e^4$
1D model:PIFNN	[200] × 3	$N_r = N_u = N_{u_t} = 240$	240	Adaptive loss weights
Layered model: PINN	[100] × 6	$N_{ic} = 1e^4, N_{pde} = 4e^4$		$\lambda_{pde} = 1, \lambda_{ic1} = \lambda_{ic2} = 1e^4$
Layered model: PINN	[100] × 6	$N_{ic} = 1e^4, N_{pde} = 4e^4$	200	Adaptive loss weights
Layered model: aPINN	[100] × 6	$N_{ic} = 1e^4, N_{pde} = 4e^4$	n = 10	$\lambda_{pde} = 1, \lambda_{ic1} = \lambda_{ic2} = 1e^4$
Layered model: PIFNN	[100] × 6	$N_{ic} = 1e^4, N_{pde} = 4e^4$		$\lambda_{pde} = 1, \lambda_{ic1} = \lambda_{ic2} = [1e^2, 5e^2, 1e^3, 3e^3, 5e^3, 7e^3, 1e^4, 5e^4, 1e^5, 1e^6]$
Layered model: PIFNN	[100] × 6	$N_{ic} = 1e^4, N_{pde} = 4e^4$	200	Adaptive loss weights

Appendix B

In some cases of this paper, we use the method of maximum normalization of the latent physical field u and scale the wave velocity to improve the training efficiency of the neural network. For the original wave equation:

$$\frac{\partial^2 u}{\partial t^2} = c^2 \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right),$$

Scale parameters are defined as $\alpha = 1/\max(c)$, $\beta = 1/\max(|u_i|)$, The equivalent transformation of the above equation is:

$$\frac{\partial^2 (\beta u)}{\partial t^2} = \alpha^2 c^2 \left(\frac{\partial^2 (\beta u)}{\partial (\alpha x)^2} + \frac{\partial^2 (\beta u)}{\partial (\alpha x)^2} \right).$$

Data availability

Data will be made available on request.

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